

Clinical Risk Assessment Report: Patient 80

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 8.9% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.40x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.2%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -2.09 [Bottom 3%] (-2.2%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Patient Age (Age) **Value: 68.0 [Avg (68th %ile)] (-1.4%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.8 [Bottom 3%] (-0.9%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 0.03 [Top 23%] (-0.9%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -0.9 [Bottom 15%] (-0.8%)**

Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.38 [Avg (26th %ile)] (-0.8%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.43 [Bottom <1%] (-0.7%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 0.5 [Avg (70th %ile)] (-0.6%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

DISCLAIMER: This report is generated by an investigational machine learning model. It is intended to provide adjunctive biological insights and must not replace standard clinical, radiological, or pathological evaluation.

Machine Learning Feature Attribution (SHAP Decision Path)

